GitHub + Frontend Setup

Quick revision notes for the GeM Bid Compliance capstone project

Project: AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement
Your focus: Data Science / AI work
Goal: Keep your work separate from the original repository while learning how to run the frontend.

1. Key GitHub concepts

Concept	Meaning
Clone	Downloads a repository onto your laptop.
Fork	Creates your own GitHub copy of another person's repository.
Branch	A separate line of development. Your branch is `amandeep-data-science`.
Push	Uploads committed changes to GitHub.

2. Our repository setup

origin = your fork on GitHub (where you push your branch)
upstream = Vanshika's original repository (kept as the reference/original project).

Commands used:

```
git remote rename origin upstream
git remote add origin https://github.com/amandeep-singh28/AI-Powered-Integrated-Bid-Complia
nce-Verification-Platform-For-GeM-Procurement.git
git remote -v
```

Your current working branch is `amandeep-data-science`. The branch was successfully pushed to your own fork and tracks `origin/amandeep-data-science`.

3. Everyday Git commands

Command	Purpose
`git status`	Shows changed, staged, and untracked files.
`git branch`	Lists local branches.
`git switch amandeep-data-science`	Switches to your working branch.
`git add .`	Stages changes; it does not upload them.
`git commit -m "message"`	Saves a checkpoint in local Git history.
`git push`	Uploads committed changes to GitHub.
`git pull`	Fetches and integrates changes from the tracked remote branch.
`git remote -v`	Shows the remote repository URLs.

Remember: Write code → `git add` → `git commit` → `git push`.

4. Running the React frontend

The project's `package.json` is inside the `frontend` folder, not the repository root. Run the commands from the repository root as follows:

```
cd frontend
npm install
npm run dev
```

Node.js: runs JavaScript tooling on your computer. **npm:** installs packages. **package.json:** lists dependencies and scripts. **Vite:** starts the local development server. Open the local URL shown in the terminal (often `http://localhost:5173/`). Keep the terminal running while using the site.

5. What we completed

• Verified Node.js and npm were available.

• Installed the frontend dependencies from the frontend folder.

• Created a fork under your GitHub account.

• Created and switched to amandeep-data-science.

• Configured origin (your fork) and upstream (Vanshika's repository).

• Successfully pushed your branch to your fork.

• Removed the accidental root-level package-lock.json and restored the frontend lockfile to avoid committing setup-related changes.

6. Useful reminders

• The fork and branch let you develop without directly changing Vanshika's original main branch.

• git add . only stages changes; git commit saves them locally; git push uploads committed changes.

• If you change dependencies later, review package.json and package-lock.json before committing.

• The npm audit result was zero vulnerabilities after npm audit fix at that time; it is not a guarantee against all vulnerabilities.

Keep this PDF for later revision. You do not need to master every concept today.